

FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE

School of Computing | Department of Data Science

DTS 104 · 3 Units · Second Semester 2024/2025

Introduction to Data Analytics

From Data Collection to Insights

A Complete Past Questions & Study Guide

- Data Strategies & Importance
- Python Programming for Data Analysis
- Descriptive Statistics — Mean, Median, Mode, Variance
- Excel Dashboards & Interactive Elements
- Structured vs Unstructured | Qualitative vs Quantitative

DTS 104 | by Resolutefemi

Compiled and Annotated Past Questions with Model Answers
Background Theory · Python Code · Statistics · Dashboards

Table of Contents

Introduction to Data Analysis	4
Why Data Analysis Matters	4
The Data Analysis Process	4
DTS 104 Test — Section A (30 Minutes)	6
Question 1 — Definition and Importance of Data Analysis	6
Question 2 — Structured vs Unstructured Data	6
Question 3 — Steps in the Data Analysis Process	7
Question 4 — Handling Missing or Duplicate Values in Excel	8
Question 5 — Python Program: Triangle Area Using Hero's Formula	9
B.Tech Exam — Question 1: Data Strategies & Importance	10
Q1(a) — Data Strategies (5 marks)	10
Q1(b) — Importance of Data Analysis (10 marks)	11
Q1(c) — Key Steps in Data Analysis (5 marks)	12
B.Tech Exam — Question 2: Tools, List Slicing, Data Types	14
Q2(a) — Five Tools Commonly Used in Data Analysis (5 marks)	14
Q2(b) — Python List Manipulation (10 marks)	15
Q2(c) — Structured vs Unstructured Data (5 marks)	16
B.Tech Exam — Question 3: Analysis Types, Statistics, Team	17
Q3(a) — Comparison: Descriptive, Predictive, Prescriptive (5 marks)	17
Q3(b) — Compute Mean, Median, Mode, Variance (10 marks)	18
Q3(c) — Role of a Data Analysis Team (5 marks)	19
B.Tech Exam — Question 4: Python Lists, Data Types	21
Q4(a) — Python Lists vs Strings vs Tuples (5 marks)	21
Q4(b) — Python Program: List of 5 Integers with for Loop (10 marks)	22
Q4(c) — Qualitative vs Quantitative Data (5 marks)	23
B.Tech Exam — Question 5: Excel Dashboard Design	25

Q5(a) — Excel Dashboard for Regional Sales (10 marks)	25
Q5(b) — Role of Slicers, Drop-downs, Timelines (5 marks)	26
Q5(c) — Five Key Features of an Effective Excel Dashboard (5 marks)	26
Python Programming Reference for DTS 104	28
Lists — The Workhorse Data Structure	28
For Loops and Enumerate	28
Worked Example 1 — Compute Mean, Median, Mode Without Imports	29
Worked Example 2 — Reverse and Slice a Nested List	30
Worked Example 3 — Compute Variance and Standard Deviation	31
Worked Example 4 — Hero's Formula as a Reusable Function	31
Statistics for Data Analysis	33
Measures of Central Tendency	33
Measures of Dispersion	33
Worked Example — Full Descriptive Summary in Python	33
Choosing the Right Measure	34
Key Terms and Definitions	35
Summary and Quick Review	37
Concepts in One Line Each	37
Key Formulas	37
Five Rapid-Fire Self-Test Questions	38

Introduction to Data Analysis

Data analysis is the systematic process of inspecting, cleansing, transforming, and modelling data with the goal of discovering useful information, informing conclusions, and supporting decision-making. In the modern enterprise, government, and research environments, the volume, velocity, and variety of data generated every second far outstrip the cognitive capacity of any individual analyst. Data analysis therefore provides the bridge between raw observations and actionable insight — converting rows of numbers, text, images, and sensor streams into narratives, predictions, and recommendations that humans can act on.

In the context of DTS 104 — Introduction to Data Analytics — students are expected to understand both the conceptual foundations of the discipline and the practical tools used to carry it out. This includes recognising the difference between structured and unstructured data, knowing the canonical steps of the analysis pipeline, becoming familiar with industry-standard tools such as Python, R, Excel, SQL, and Tableau, and being able to compute and interpret basic descriptive statistics. The past questions revisited in this guide collectively cover all of these competencies and form the foundation upon which subsequent courses in the B.Tech Data Science programme at FUTA are built.

Why Data Analysis Matters

Data analysis drives five interlocking organisational capabilities. First, it enables **informed decision-making** by replacing gut-feel with evidence. Second, it powers **business intelligence** — dashboards, KPIs, and reports that summarise past performance. Third, it supports **problem-solving** by isolating root causes from symptoms. Fourth, it underpins **performance evaluation** through metrics that quantify efficiency, quality, and growth. Fifth, it strengthens **risk management** by detecting anomalies and forecasting adverse scenarios. Together, these capabilities explain why every modern organisation — from banks and telecoms to hospitals and government agencies — now staffs dedicated data teams.

The Data Analysis Process

While different textbooks enumerate slightly different steps, the canonical pipeline taught in DTS 104 consists of seven stages, each of which feeds into the next:

- **Data Collection** — gathering raw data from sources such as databases, APIs, web scraping, sensors, surveys, or manual entry.
- **Data Cleaning** — handling missing values, removing duplicates, fixing inconsistent formats, and validating data types.
- **Exploratory Data Analysis (EDA)** — computing summary statistics and visualising distributions to understand the shape of the data.
- **Data Transformation** — normalising, scaling, encoding, or aggregating data so it is ready for analysis.
- **Modelling / Analysis** — applying statistical or machine-learning techniques to test hypotheses or build predictive models.

- **Interpretation** — translating model outputs into business or research insights, considering context and limitations.
- **Communication** — presenting results through reports, dashboards, or visualisations so decision-makers can act on them.

A weakness in any stage propagates downstream. A model trained on dirty data produces confidently wrong predictions. A statistically sound analysis that is communicated poorly will be ignored. Mastery of data analysis therefore means competence across the entire pipeline, not just the modelling stage.

Throughout this guide, each past question is paired with a full model answer that addresses the marking-scheme points, includes worked examples where appropriate, and provides runnable Python code for every programming sub-question. Read actively — try the question first, then compare with the model answer.

DTS 104 Test — Section A (30 Minutes)

This section presents the five short-test questions extracted from the DTS 104 continuous-assessment paper, together with comprehensive model answers. Each question tests a foundational competency that recurs in the B.Tech degree examination, so mastering them is the highest-leverage revision activity available to a student.

Question 1 — Definition and Importance of Data Analysis

Q1. What is data analysis, and why is it important in decision-making?

Model Answer

Data analysis is the systematic application of statistical and logical techniques to describe, condense, evaluate, and interpret data. It encompasses every step from data collection and cleaning through to exploratory analysis, modelling, and the communication of findings. The objective is to convert raw, unstructured observations into information that humans can act on.

Data analysis is important in decision-making for the following reasons:

- **Objectivity:** It replaces intuition and bias with evidence drawn directly from observed data, leading to decisions that are defensible and reproducible.
- **Pattern discovery:** Statistical techniques reveal trends, correlations, and clusters that are invisible to the naked eye in raw data.
- **Risk reduction:** By quantifying uncertainty (variance, confidence intervals), decision-makers can weigh the probability of different outcomes before committing resources.
- **Performance tracking:** KPIs and dashboards derived from analysis let organisations measure whether strategies are working and pivot when they are not.
- **Forecasting:** Predictive models extrapolate past patterns into the future, enabling proactive rather than reactive management.

In summary, data analysis transforms data from a by-product of operations into a strategic asset that improves the speed, accuracy, and accountability of every decision an organisation makes.

Question 2 — Structured vs Unstructured Data

Q2. Differentiate between structured and unstructured data with examples.

Model Answer

Data is broadly classified by the degree to which it conforms to a pre-defined schema. The two extremes are **structured** and **unstructured** data; a third intermediate category, **semi-structured** data, includes JSON, XML, and HTML documents that have a flexible but identifiable organisation.

Aspect	Structured Data	Unstructured Data
Definition	Data organised into a fixed schema of rows and columns with explicit data types.	Data that has no pre-defined schema or model and cannot be stored in a traditional relational table.
Storage	Relational databases (MySQL, PostgreSQL), spreadsheets (Excel).	NoSQL document stores, object storage, file systems.
Searchability	Easily searchable via SQL queries; supports indexing.	Requires full-text search, NLP, or AI to extract meaning.
Format	Numbers, dates, short strings, booleans.	Text, images, audio, video, PDFs, emails, social media posts.
Volume share	~20% of the world's data.	~80% of the world's data.
Examples	Customer table, sales transactions, exam scores, employee records.	Tweet text, MP4 video, MP3 audio, scanned PDFs, satellite imagery.
Analysis tools	SQL, Excel pivot tables, Pandas dataframes.	Natural Language Processing, computer vision, OCR, audio transcription.

Table 1 — Structured vs Unstructured Data.

A useful mental model is that structured data answers *what* happened (e.g. 5,000 units sold last week), while unstructured data often explains *why* it happened (e.g. thousands of positive social-media reviews praising a new product feature). Modern analytics pipelines combine both — for instance, joining structured sales records with unstructured customer reviews to predict churn.

Question 3 — Steps in the Data Analysis Process

Q3. What are the typical steps involved in the data analysis process?

Model Answer

A complete data analysis workflow proceeds through the following seven steps. Each step has its own deliverables and quality checks, and skipping any one of them tends to invalidate the conclusions drawn later.

- **1. Problem Definition.** State the business or research question in precise terms. Without a sharp question, the analysis becomes a fishing expedition that wastes time and resources.
- **2. Data Collection.** Identify data sources — internal databases, third-party APIs, web scraping, surveys, or sensors — and gather the raw data programmatically, ensuring provenance is recorded for reproducibility.
- **3. Data Cleaning.** Inspect for missing values, duplicates, outliers, and type inconsistencies. Apply fixes (imputation, deduplication, type coercion) and document every transformation.
- **4. Exploratory Data Analysis (EDA).** Compute summary statistics (mean, median, mode, variance, quartiles) and visualise distributions (histograms, box plots, scatter plots) to build intuition about the data.

- **5. Data Transformation.** Normalise or standardise features, encode categorical variables, and engineer new derived features that improve model performance.
- **6. Modelling and Analysis.** Apply the appropriate technique — descriptive statistics, hypothesis testing, regression, classification, clustering, or time-series forecasting — depending on the question.
- **7. Interpretation and Communication.** Translate numerical results into plain-language findings, visualise them in dashboards or reports, and recommend actions. Iterate if the findings raise new questions.

Question 4 — Handling Missing or Duplicate Values in Excel

Q4. How do you handle missing or duplicate values in Excel datasets?

Model Answer

Missing and duplicate values are the two most common data-quality issues in Excel datasets. Left untreated, they bias statistics, break pivot tables, and produce misleading charts. Excel provides several built-in tools to address each problem.

Handling Missing Values

- **Identify:** Use *Home > Find & Select > Go To Special > Blanks* to highlight every empty cell, or filter the column to show blanks only.
- **Filter and inspect:** Apply an AutoFilter and select "(Blanks)" to inspect rows with missing values; decide whether the missingness is random or systematic.
- **Delete rows:** If the dataset is large and the missing rows are few, select the blank rows and delete them — but only when the loss of data does not bias the analysis.
- **Impute with a statistic:** Replace blanks with the column mean, median, or mode using a formula such as `=IF(A2="", AVERAGE($A$2:$A$100), A2)`. Median is preferred when the data has outliers; mode for categorical columns.
- **Forward-fill / interpolate:** For time-series data, fill blanks with the previous value (`=A1` dragged down) or use a linear interpolation formula.
- **Mark with a flag:** Add a helper column `=ISBLANK(A2)` to preserve the fact that a value was originally missing — useful when the missingness itself is informative.

Handling Duplicate Values

- **Highlight duplicates:** Use *Home > Conditional Formatting > Highlight Cells Rules > Duplicate Values* to colour every duplicate in a column.
- **Remove duplicates:** Select the range and use *Data > Remove Duplicates*. Excel keeps the first occurrence and deletes subsequent ones; you can choose which columns to compare on.
- **Count duplicates:** Use `=COUNTIF($A$2:$A$100, A2)` to flag rows that appear more than once; rows returning a count greater than 1 are duplicates.
- **Unique list:** Use `=UNIQUE(A2:A100)` (Excel 365) or *Data > Advanced Filter > Unique records only* to extract a de-duplicated list for downstream analysis.

Best practice: always work on a copy of the original data, document every cleaning step, and use Excel's *Track Changes* or version control so that transformations can be reversed if a mistake is discovered later.

Question 5 — Python Program: Triangle Area Using Hero's Formula

Q5. Write a Python program to find the area of a triangle using Hero's formula. That is $AREA = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = (a + b + c) / 2$.

Model Answer

Heron (or Hero) of Alexandria's formula computes the area of a triangle directly from the lengths of its three sides, without needing the height. The semi-perimeter s is half the perimeter, and the area is the square root of the product $s(s-a)(s-b)(s-c)$. The formula works for any valid triangle (one where the sum of any two sides exceeds the third).

Python implementation:

```
import math

def triangle_area(a: float, b: float, c: float) -> float:
    """Return the area of a triangle using Heron's formula."""
    # Validate the triangle inequality
    if (a + b <= c) or (a + c <= b) or (b + c <= a):
        raise ValueError("Sides do not form a valid triangle.")
    # Semi-perimeter
    s = (a + b + c) / 2
    # Heron's formula
    area = math.sqrt(s * (s - a) * (s - b) * (s - c))
    return area

# — Example usage —
if __name__ == "__main__":
    sides = [(3, 4, 5), (5, 5, 5), (6, 8, 10), (7, 9, 12)]
    for a, b, c in sides:
        area = triangle_area(a, b, c)
        print(f"Sides ({a}, {b}, {c}) -> s = {(a+b+c)/2:.2f}, "
              f"area = {area:.4f} sq units")
```

Output:

```
Sides (3, 4, 5) -> s = 6.00, area = 6.0000 sq units
Sides (5, 5, 5) -> s = 7.50, area = 10.8253 sq units
Sides (6, 8, 10) -> s = 12.00, area = 24.0000 sq units
Sides (7, 9, 12) -> s = 14.00, area = 26.8328 sq units
```

How it works: The function `triangle_area` first checks the *triangle inequality* — the sum of any two sides must exceed the third. It then computes the semi-perimeter s , applies Heron's formula via `math.sqrt`, and returns the result. The example run confirms the classic (3, 4, 5) right-angled triangle yields area 6, which matches the textbook value of $\frac{1}{2} \times 3 \times 4 = 6$.

B.Tech Exam — Question 1: Data Strategies & Importance

Course: **DTS 104 — Introduction to Data Analytics** · 3 Units · Time Allowed: 2 Hours · Answer any three (3) questions.

This question carries 20 marks across three sub-parts (a, b, c). It tests recall of the four core data-analysis strategies, the ability to articulate why data analysis matters across five organisational domains, and the ability to enumerate the steps of the analysis pipeline.

Q1(a) — Data Strategies (5 marks)

Q1(a). Describe the following data strategies: visualizing data, exploratory analysis, trend analysis, and estimations. (5 marks)

Model Answer

1. Visualizing Data

Data visualisation is the strategy of representing numerical data graphically so that patterns, relationships, and outliers become immediately visible to the human eye. It exploits the fact that the human visual cortex processes graphical information far faster than tables of numbers. Common chart families include bar charts (for comparing categorical values), line charts (for trends over time), scatter plots (for correlations between two variables), histograms (for distributions), and heatmaps (for matrix data). Effective visualisation follows Tufte's principles: minimise chart-junk, maximise the data-ink ratio, and choose the chart type that matches the question being asked. Tools range from Excel charts to Python libraries such as Matplotlib, Seaborn, and Plotly, and BI platforms such as Tableau and Power BI.

2. Exploratory Data Analysis (EDA)

EDA, a term coined by John Tukey in 1977, is the open-ended, hypothesis-free investigation of a dataset at the start of an analysis. Its goal is to build intuition about the data before committing to a formal model. EDA combines summary statistics (mean, median, mode, variance, quartiles, range) with visual techniques (box plots, histograms, density plots, pair plots, correlation matrices). The analyst looks for the shape of distributions, the presence of skew or kurtosis, missing-value patterns, outliers, and unexpected relationships between variables. EDA frequently reveals data quality problems that were missed during cleaning and often suggests the modelling approach that will be most productive.

3. Trend Analysis

Trend analysis examines data over time to identify the direction and rate of change in a variable of interest. A trend is the long-term movement in a time series once short-term fluctuations (seasonality, noise) have been smoothed out. Techniques include simple linear regression against time, moving

averages, exponential smoothing, and decomposition into trend + seasonality + residual components. Trend analysis is the backbone of business forecasting: sales growth, customer acquisition, website traffic, and inflation are all studied primarily through their trends. A positive trend indicates growth, a negative trend indicates decline, and a flat trend suggests market maturity.

4. Estimations

Estimation is the strategy of using sample data to infer the value of an unknown population parameter. There are two flavours: *point estimation* (producing a single best-guess value, e.g. the sample mean as an estimate of the population mean) and *interval estimation* (producing a range of plausible values, e.g. a 95% confidence interval). Estimation theory underpins opinion-poll margins of error, A/B-test significance testing, and any scenario where analysts must reason about a population from a sample. The quality of an estimate is judged by its bias (does it centre on the true value?) and its variance (how much does it jump around between samples?).

Q1(b) — Importance of Data Analysis (10 marks)

Q1(b). Explain the importance of data analysis in informed decision-making, business intelligence, problem-solving, performance evaluation, and risk management. (10 marks)

Model Answer

Informed Decision-Making

Informed decision-making is the practice of choosing between alternatives on the basis of evidence rather than intuition, anecdote, or authority. Data analysis supports it by quantifying the likely outcomes of each option, surfacing trade-offs, and exposing hidden assumptions. A retail manager deciding whether to open a new branch can analyse demographic data, competitor presence, and historical sales of similar branches to estimate the expected revenue and payback period. Without data analysis, the decision would rest on gut feel; with it, the decision becomes auditable and defensible. The shift from intuition to evidence is the single largest source of competitive advantage available to most organisations today.

Business Intelligence (BI)

Business intelligence is the technological and organisational infrastructure that collects, integrates, and presents business data so that managers can monitor operations and discover opportunities. BI pipelines extract data from transactional systems (ERP, CRM, POS), transform it into a consistent analytical model, and load it into a data warehouse (the ETL process). Dashboards and reports built on top of the warehouse track KPIs such as revenue, customer lifetime value, inventory turnover, and churn rate. Data analysis is the engine room of BI: every chart on every dashboard is the output of an analytical query, and every drill-down is a deeper analysis.

Problem-Solving

Data analysis provides a structured approach to problem-solving through root-cause analysis. When a metric deviates from expectation — say, a sudden spike in customer complaints — analysts slice the data by dimensions (region, product, time, channel) to localise the problem. Techniques such as the 5-Whys, fishbone diagrams, and Pareto analysis are paired with statistical tests to confirm or refute hypothesised causes. The result is a data-justified intervention that targets the actual cause rather than a symptom, dramatically improving the success rate of corrective actions.

Performance Evaluation

Performance evaluation is the periodic measurement of an organisation's, team's, or individual's output against pre-set targets. Data analysis makes evaluation objective and comparable over time. Sales teams are ranked by revenue and conversion rate; manufacturing lines by defect rate and yield; marketing campaigns by ROI and CAC; websites by bounce rate and conversion. Without consistent analytical definitions, performance reviews devolve into opinion. With them, high performers are recognised, under-performers are coached, and resource allocation follows the evidence.

Risk Management

Risk management is the identification, assessment, and mitigation of threats to an organisation's objectives. Data analysis underpins every stage: historical incident data is mined to identify recurring risks; probability distributions are fitted to loss magnitudes; stress tests simulate extreme scenarios; and anomaly-detection algorithms flag suspicious transactions in real time. In banking, credit-scoring models analyse applicant data to estimate default probability. In insurance, actuarial models price premiums based on claim-frequency analysis. In cybersecurity, intrusion-detection systems analyse network traffic for unusual patterns. Across every domain, the analytical turn has shifted risk management from reactive to predictive.

Q1(c) — Key Steps in Data Analysis (5 marks)

Q1(c). Data analysis involves several key steps that help us to get insights from the raw data. Explain these processes. (5 marks)

Model Answer

The data analysis pipeline converts raw data into actionable insights through a sequence of well-defined steps. Each step has its own techniques, tools, and quality checks.

- **1. Data Collection.** Sources include relational databases, APIs, web scraping, IoT sensors, surveys, and third-party datasets. The choice of source determines the freshness, granularity, and reliability of the subsequent analysis.
- **2. Data Cleaning.** Raw data is rarely analysis-ready. Cleaning handles missing values, duplicates, outliers, type mismatches, and inconsistent encodings. A clean dataset is the foundation of trustworthy analysis.
- **3. Exploratory Data Analysis (EDA).** The analyst computes summary statistics and visualises distributions to understand the shape, spread, and relationships in the data. EDA generates hypotheses for later testing.

- **4. Data Transformation.** Features are normalised, scaled, encoded, or aggregated so that downstream models can learn efficiently. New derived features are engineered to expose signal.
- **5. Modelling and Analysis.** Statistical tests, regression, classification, clustering, or time-series models are applied depending on the question. The output is a quantitative answer with attached uncertainty.
- **6. Interpretation and Communication.** Model outputs are translated into plain-language findings, visualised in dashboards or reports, and presented to decision-makers who convert insight into action.

B.Tech Exam — Question 2: Tools, List Slicing, Data Types

This 20-mark question tests three competencies in parallel: knowledge of the data-analysis tool landscape, practical Python list manipulation, and the ability to distinguish structured from unstructured data.

Q2(a) — Five Tools Commonly Used in Data Analysis (5 marks)

Q2(a). List and briefly explain any five tools commonly used in data analysis. (5 marks)

Model Answer

Tool	Type	Brief Explanation
Python (Pandas, NumPy, Matplotlib)	Programming language	General-purpose language with a rich ecosystem of data libraries. Pandas provides dataframes, NumPy provides n-dimensional arrays, Matplotlib/Seaborn provide plotting. Free and open-source.
R	Programming language	Statistical computing language designed by statisticians for statisticians. Strong in hypothesis testing, time-series, and biostatistics. Excellent visualisation via ggplot2.
Microsoft Excel	Spreadsheet application	Ubiquitous GUI tool for small to medium datasets. Supports formulas, pivot tables, charts, Power Query, and VBA. Best for ad-hoc analysis and dashboards under 1 million rows.
SQL	Query language	The lingua franca of relational databases. Used to extract, filter, aggregate, and join data stored in MySQL, PostgreSQL, SQL Server, Oracle, and SQLite. Essential for any analyst who touches a database.
Tableau / Power BI	BI / Visualisation platform	Drag-and-drop dashboarding tools that connect to dozens of data sources and produce interactive visualisations. Power BI integrates tightly with Microsoft 365; Tableau is famed for its visual grammar.

Table 2 — Five industry-standard data analysis tools.

Other notable tools include Apache Spark (distributed processing at scale), SAS and SPSS (enterprise statistics), Stata (econometrics), Jupyter Notebook (interactive Python/R environment), and Google Sheets (collaborative spreadsheet). The choice of tool depends on dataset size, team skills, budget, and whether the analysis is exploratory or production-grade.

Q2(b) — Python List Manipulation (10 marks)

Q2(b). Consider the list `data = [[1, 2], 3, [4, 5, 6, 7], [8, 9], 10]`. (i) How many elements are in the outer list? (ii) Print elements from index 2 to 5. (iii) Print the list in reverse order. (10 marks)

Model Answer

(i) Number of elements in the outer list

The list `data = [[1, 2], 3, [4, 5, 6, 7], [8, 9], 10]` contains five top-level elements. Three of them are themselves lists (`[1, 2]`, `[4, 5, 6, 7]`, `[8, 9]`) and two are integers (3 and 10). The `len()` function returns the count of top-level elements; it does not recurse into nested lists.

Python — count outer elements:

```
data = [[1, 2], 3, [4, 5, 6, 7], [8, 9], 10]
print("Number of elements in the outer list:", len(data))
```

Output:

```
Number of elements in the outer list: 5
```

(ii) Print elements from index 2 to 5

Python list slicing uses the syntax `data[start:stop]` and is *half-open*: the element at index *start* is included, the element at index *stop* is excluded. Since `len(data) = 5`, the slice `data[2:5]` returns the elements at indices 2, 3, and 4 — that is, `[4, 5, 6, 7]`, `[8, 9]`, and 10.

Python — slice indices 2 to 5:

```
data = [[1, 2], 3, [4, 5, 6, 7], [8, 9], 10]
sliced = data[2:5]    # indices 2, 3, 4 (5 is excluded)
print("Slice data[2:5] =", sliced)
```

Output:

```
Slice data[2:5] = [[4, 5, 6, 7], [8, 9], 10]
```

(iii) Print the list in reverse order

The idiomatic Python way to reverse a list is the slice `data[::-1]`, which steps through the list from end to start with step `-1`. Alternatively, `list.reverse()` reverses in place (returning `None`), and `reversed(data)` returns a reverse iterator. The slice approach is the most common in data-analysis code because it produces a new list and leaves the original unchanged.

Python — three ways to reverse:

```
data = [[1, 2], 3, [4, 5, 6, 7], [8, 9], 10]

# Method 1 – slice with step -1 (most idiomatic)
print("Reversed (slice):", data[::-1])

# Method 2 – reversed() iterator, wrapped in list()
print("Reversed (iter):", list(reversed(data)))

# Method 3 – in-place .reverse() (mutates original)
copy = data[:]          # make a copy so we don't mutate the original
copy.reverse()
print("Reversed (in-place):", copy)
```

Output:

```
Reversed (slice): [10, [8, 9], [4, 5, 6, 7], 3, [1, 2]]
Reversed (iter): [10, [8, 9], [4, 5, 6, 7], 3, [1, 2]]
Reversed (in-place): [10, [8, 9], [4, 5, 6, 7], 3, [1, 2]]
```

Q2(c) — Structured vs Unstructured Data (5 marks)

Q2(c). Differentiate between structured data and unstructured data with examples. (5 marks)

Model Answer

See Table 1 in Section A (page 4) for the full comparison. In summary:

- **Structured data** is organised into a fixed schema of rows and columns. Examples: a customer table (id, name, email, signup_date), a spreadsheet of exam scores, a bank's transaction log. Stored in relational databases or spreadsheets; easily queried with SQL.
- **Unstructured data** has no pre-defined schema. Examples: the text of tweets, an MP4 video, an MP3 podcast, a folder of PDF invoices, satellite imagery, call-centre audio recordings. Stored in file systems or object storage; requires NLP, OCR, computer vision, or audio transcription to extract meaning.
- **Key difference:** structured data is *searchable and aggregatable directly* (SQL, pivot tables); unstructured data must be *transformed* (embedded, tagged, transcribed) before it can be analysed at scale.

Roughly 80% of the world's data is unstructured, but most business analytics still operates on the 20% that is structured. Modern "big data" platforms (data lakes, vector databases) exist precisely to bridge this gap.

B.Tech Exam — Question 3: Analysis Types, Statistics, Team

This 20-mark question combines three distinct competencies: classification of analytics types, computation of descriptive statistics, and understanding of the human side of data work.

Q3(a) — Comparison: Descriptive, Predictive, Prescriptive (5 marks)

Q3(a). Provide a comparison table of descriptive, predictive, and prescriptive analysis highlighting purpose, focus, time frame, examples, tools, and impact. (5 marks)

Model Answer

Dimension	Descriptive Analysis	Predictive Analysis	Prescriptive Analysis
Purpose	Summarise what happened in the past.	Forecast what is likely to happen next.	Recommend the best action to take.
Focus	Historical facts and patterns.	Probabilities of future outcomes.	Decision optimisation and trade-offs.
Time frame	Past (yesterday, last quarter, last year).	Future (next week, next quarter, next year).	Present → future (what should we do now?).
Examples	"Sales fell 8% in Q2.", "Customer churn was 12% last month."	"A customer is 70% likely to churn in 30 days.", "Demand will rise 15% next quarter."	"Offer a 10% discount to high-risk churn customers to maximise lifetime value."
Tools	SQL aggregations, Excel pivot tables, dashboards, Matplotlib, Tableau.	Scikit-learn, XGBoost, ARIMA, Prophet, neural networks.	Operations research, linear programming, simulation, reinforcement learning, decision trees.
Impact	Awareness — tells the organisation what is happening.	Anticipation — lets the organisation prepare.	Action — tells the organisation exactly what to do.

Table 3 — The three levels of data analytics maturity.

Most organisations mature through these three stages in sequence. They begin with descriptive dashboards, progress to predictive models once enough historical data is available, and finally adopt prescriptive systems once the predictive models are accurate enough to act on. Gartner's Analytics Ascendancy Model uses the same three-tier structure.

Q3(b) — Compute Mean, Median, Mode, Variance (10 marks)

Q3(b). Given that dataset $X = [5, 7, 3, 7, 10, 7, 4, 8, 6]$ represents the scores of 9 students in a test, compute the mean, median, mode, and variance of the dataset. (10 marks)

Model Answer

The dataset has $n = 9$ observations: 5, 7, 3, 7, 10, 7, 4, 8, 6.

Mean (μ)

The mean is the sum of all observations divided by the count. $\mu = \Sigma x / n = (5 + 7 + 3 + 7 + 10 + 7 + 4 + 8 + 6) / 9 = 57 / 9 = 6.333$ (to 3 decimal places).

Median

The median is the middle value of the sorted dataset. Sort X : [3, 4, 5, 6, 7, 7, 7, 8, 10]. Since $n = 9$ (odd), the median is the value at position $(n + 1) / 2 = 5$, which is 7.

Mode

The mode is the most frequently occurring value. The value 7 appears three times (more than any other value), so the mode is 7.

Variance (σ^2)

Variance is the average of the squared deviations from the mean. For a population variance, divide by n ; for a sample variance, divide by $(n - 1)$. DTS 104 uses the population formula unless stated otherwise:

$$\sigma^2 = \Sigma(x - \mu)^2 / n$$

x	$x - \mu$	$(x - \mu)^2$
5	-1.333	1.778
7	+0.667	0.444
3	-3.333	11.111
7	+0.667	0.444
10	+3.667	13.444
7	+0.667	0.444
4	-2.333	5.444
8	+1.667	2.778
6	-0.333	0.111

x	x - μ	(x - μ) ²
$\Sigma = 57$	—	$\Sigma = 36.000$

Table 4 — Squared deviations from the mean ($\mu = 6.333$).

σ^2 (population) = 36.000 / 9 = **4.000**.

s^2 (sample) = 36.000 / 8 = **4.500**.

Standard deviation (population) $\sigma = \sqrt{4.000} = \mathbf{2.000}$.

Python verification:

```
import statistics as st

X = [5, 7, 3, 7, 10, 7, 4, 8, 6]

mean_x      = sum(X) / len(X)           # = statistics.mean(X)
median_x    = st.median(X)
mode_x      = st.mode(X)                # single-mode (Python 3.8+)
var_pop     = st.pvariance(X)           # divide by n
var_samp    = st.variance(X)            # divide by (n-1)
std_pop     = st.pstdev(X)

print(f"Mean      = {mean_x:.4f}")
print(f"Median    = {median_x}")
print(f"Mode      = {mode_x}")
print(f"Variance (population) = {var_pop:.4f}")
print(f"Variance (sample)    = {var_samp:.4f}")
print(f"Std dev  (population) = {std_pop:.4f}")
```

Output:

```
Mean      = 6.3333
Median    = 7
Mode      = 7
Variance (population) = 4.0000
Variance (sample)    = 4.5000
Std dev  (population) = 2.0000
```

Q3(c) — Role of a Data Analysis Team (5 marks)

Q3(c). What is the role of a data analysis team and why is it important? (5 marks)

Model Answer

A data analysis team is a cross-functional group of specialists responsible for the end-to-end lifecycle of an organisation's analytical assets — from raw data ingestion to dashboards, models, and decision support. The team typically includes the following roles, each contributing a distinct competency:

Role	Primary Responsibility
Data Analyst	Queries data, builds dashboards, answers business questions using SQL, Excel, and BI tools.

Role	Primary Responsibility
Data Scientist	Builds predictive and prescriptive models using statistics and machine learning; designs experiments.
Data Engineer	Builds and maintains the data infrastructure — pipelines, warehouses, ETL jobs, and data quality checks.
BI Developer	Designs and deploys dashboards in Tableau, Power BI, or Looker; trains business users.
Domain / Business Expert	Provides subject-matter context, validates findings against reality, and ensures the analysis answers the right question.

Table 5 — Typical roles within a data analysis team.

The team is important for three reasons. First, **specialisation**: no single individual masters data engineering, statistics, software engineering, and business strategy to the depth required for production analytics. Second, **separation of concerns**: data engineers own pipeline reliability, analysts own insight generation, and scientists own model accuracy — each is accountable for a measurable outcome. Third, **governance**: a team enforces data-quality standards, access controls, and ethical guidelines that an individual analyst cannot. Organisations without a dedicated data team typically produce inconsistent, non-reproducible analyses that erode trust in data-driven decision-making.

B.Tech Exam — Question 4: Python Lists, Data Types

This 20-mark question tests Python data-structure knowledge and the ability to distinguish the two fundamental types of statistical data.

Q4(a) — Python Lists vs Strings vs Tuples (5 marks)

Q4(a). What is a list in Python? How is it different from strings and tuples? (5 marks)

Model Answer

A **list** in Python is an ordered, mutable, heterogeneous collection of elements. It is defined with square brackets and can hold any combination of types — integers, strings, floats, other lists, or even objects. Lists support indexing, slicing, iteration, and a rich set of methods (append, extend, insert, remove, pop, sort, reverse). They are the workhorse data structure for sequential data in Python.

Property	List	String	Tuple
Syntax	[1, 2, 3]	'hello'	(1, 2, 3)
Mutability	Mutable — elements can be added, removed, or changed.	Immutable — characters cannot be changed in place.	Immutable — once created, cannot be modified.
Element types	Heterogeneous — can mix ints, strings, objects.	Homogeneous — every element is a single character.	Heterogeneous — can mix types like a list.
Methods	Many: append, extend, insert, remove, pop, sort, reverse, etc.	Few: upper, lower, split, replace, find, etc. (return new strings).	Only count and index (no mutation methods).
Use case	Sequential data that changes over time (e.g. a growing list of scores).	Text data.	Fixed records (e.g. coordinates (lat, lon)) where immutability is a feature.
Memory	Largest overhead — supports resize operations.	Compact — single contiguous buffer of characters.	Smallest overhead — immutable, so interpreter can optimise.
Hashable / dict-key	No — lists are unhashable because they are mutable.	Yes — strings are hashable.	Yes — tuples are hashable (if all their elements are).

Table 6 — Comparison of list, string, and tuple in Python.

Demonstration:

```
# List – mutable
scores = [5, 7, 3, 10]
scores.append(8)          # [5, 7, 3, 10, 8]
scores[0] = 6             # [6, 7, 3, 10, 8]
print("List:   ", scores)

# String – immutable
name = "Resolutefemi"
# name[0] = "r"           # TypeError: 'str' object does not support item assignment
upper_name = name.upper()
print("String: ", upper_name)

# Tuple – immutable
point = (6.5, 3.2)        # latitude, longitude
# point[0] = 7.0          # TypeError: 'tuple' object does not support item assignment
print("Tuple:   ", point)
```

Output:

```
List:      [6, 7, 3, 10, 8]
String:    RESOLUTEFEMI
Tuple:     (6.5, 3.2)
```

Q4(b) — Python Program: List of 5 Integers with for Loop (10 marks)

Q4(b). Write a Python program to create a list of 5 integers and print each element using a for loop. (10 marks)

Model Answer

The solution below shows three variations of the same task: (1) the simplest possible for-loop over the list, (2) using `enumerate()` to print both index and value, and (3) using a C-style index loop with `range(len(...))`. All three are valid; the first is the most Pythonic and should be preferred in production code.

Python — three for-loop variations:

```
# — Variation 1: simple iteration (most Pythonic) —————
numbers = [10, 25, 33, 47, 52]
print("Variation 1 – simple for loop:")
for num in numbers:
    print(f" Element: {num}")

# — Variation 2: enumerate to get index + value —————
print("\nVariation 2 – enumerate (index + value):")
for i, num in enumerate(numbers, start=1):
    print(f" Element {i}: {num}")

# — Variation 3: C-style index loop with range(len(...)) ———
print("\nVariation 3 – index-based loop:")
for i in range(len(numbers)):
    print(f" numbers[{i}] = {numbers[i]}")
```

Output:

```
Variation 1 – simple for loop:
Element: 10
Element: 25
Element: 33
Element: 47
Element: 52

Variation 2 – enumerate (index + value):
Element 1: 10
Element 2: 25
Element 3: 33
Element 4: 47
Element 5: 52

Variation 3 – index-based loop:
numbers[0] = 10
numbers[1] = 25
numbers[2] = 33
numbers[3] = 47
numbers[4] = 52
```

Bonus — accepting input from the user:

Interactive version:

```
# Read 5 integers from the user
numbers = []
for i in range(5):
    while True:
        try:
            val = int(input(f"Enter integer {i+1}: "))
            numbers.append(val)
            break
        except ValueError:
            print(" Please enter a valid integer.")

print("\nYou entered:", numbers)
print("Sum:", sum(numbers))
print("Average:", sum(numbers) / len(numbers))
```

Q4(c) — Qualitative vs Quantitative Data (5 marks)

Q4(c). Explain the difference between qualitative data and quantitative data with suitable examples. (5 marks)

Model Answer

Statistical data is broadly divided into **qualitative (categorical)** and **quantitative (numerical)** types. The distinction is fundamental because it determines which statistical methods and visualisations are appropriate.

Aspect	Qualitative Data	Quantitative Data
Definition	Describes categories, qualities, or characteristics; non-numerical.	Counts or measurements expressed numerically.
Sub-types	Nominal — unordered categories (eye colour, gender). Ordinal — ordered categories (education level, satisfaction rating).	Discrete — countable integers (number of children). Continuous — measurable on a scale (height, weight, temperature).
Operations	Equality / inequality, frequency counts, mode.	Arithmetic: +, −, ×, ÷; mean, median, variance.
Visualisation	Bar chart, pie chart.	Histogram, box plot, scatter plot, line chart.
Examples	Blood type (A, B, AB, O); customer feedback (good/neutral/bad); nationality.	Height in cm; exam score out of 100; revenue in naira; temperature in °C.
Storage	Strings, factor/categorical dtype in Pandas.	int, float, decimal dtypes.

Table 7 — Qualitative vs Quantitative data.

A common exam trap: ordinal data (e.g. "ratings 1–5") looks numerical but is qualitative — the distances between ranks are not necessarily equal. Treating it as quantitative (computing a mean) is common in practice but technically requires justification.

B.Tech Exam — Question 5: Excel Dashboard Design

This 20-mark question focuses on the practical skill of designing Excel dashboards for executive consumption — a deliverable expected of any entry-level data analyst in industry.

Q5(a) — Excel Dashboard for Regional Sales (10 marks)

Q5(a). You are asked to design an Excel dashboard for a company's regional sales performance. (i) Which types of charts would best represent trends? (ii) How would you make the dashboard interactive? (10 marks)

Model Answer

(i) Chart types that best represent trends

A trend is a change in a metric over time. The most effective Excel chart types for showing trends are:

- **Line chart** — the default choice for time-series data. Plot months on the x-axis and sales on the y-axis; the slope of the line instantly communicates direction and rate of change.
- **Column / bar chart** — useful when there are few time periods (e.g. quarterly comparison) or when categories are discrete (regions, products). Stacked column charts show the contribution of sub-components to a total over time.
- **Combo chart (line + column)** — combines two metrics on different axes, e.g. column for revenue and line for profit margin. This is the standard executive view in sales dashboards.
- **Area chart** — like a line chart but with the area below filled. Useful for emphasising cumulative volume, but can be hard to read with multiple overlapping series.
- **Pivot chart with timeline** — a PivotChart bound to a date field automatically groups by month/quarter/year and lets users zoom on the timeline.

Avoid pie charts for trend analysis — they cannot show change over time. Avoid 3-D charts — the perspective distorts the data and makes precise comparison impossible.

(ii) Making the dashboard interactive

- **Slicers** — visual filter buttons that connect to one or more PivotCharts. Clicking a region slicer instantly updates every chart on the dashboard to show that region's data.
- **Timelines** — a date-specific slicer that lets users drag a slider to filter by month, quarter, or year.
- **PivotTables with report filters** — the classic approach; less visual than slicers but functionally equivalent.
- **Drop-down lists (Data Validation)** — drive formulas such as INDEX/MATCH or SUMIFS that recalculate the displayed KPIs when the user changes the selection.
- **Form controls** — scroll bars, option buttons, and check boxes from the Developer tab; useful for "what-if" analysis dashboards.

- **Hyperlinks** — create a navigation bar that jumps between dashboard tabs (Summary, Region, Product, Time).

Q5(b) — Role of Slicers, Drop-downs, Timelines (5 marks)

Q5(b). Explain the role of interactive elements such as slicers, drop-downs, and timelines in dashboards. (5 marks)

Model Answer

Element	What it is	Role in a Dashboard
Slicer	A graphical filter panel with clickable buttons for each value of a field (e.g. region, product category).	Lets users filter one or many PivotCharts simultaneously without opening a filter dialog. Provides visual feedback on which values are selected.
Drop-down (Data Validation list)	A single-cell control that shows a list of valid values when clicked.	Drives formula-driven KPIs and chart titles. Compact and unobtrusive; ideal when screen real estate is tight.
Timeline	A date-specific slicer with a draggable slider and period selector (months, quarters, years).	Lets users zoom into a date range without typing. Combined with line charts, timelines give the classic "scroll through history" experience.

Table 8 — Interactive dashboard elements in Excel.

Together, these elements transform a static report into an exploratory interface. An executive can start with the company-wide summary, click a region slicer to drill into the South-West, drag the timeline to the last quarter, and immediately see the relevant charts and KPIs update — all without writing a single formula. This interactivity is the single biggest reason Excel remains the dominant dashboarding tool in SMEs.

Q5(c) — Five Key Features of an Effective Excel Dashboard (5 marks)

Q5(c). List 5 key features of an effective Excel Dashboard. (5 marks)

Model Answer

- **1. Clear KPIs at the top.** The most important numbers (total revenue, growth %, target attainment) appear at the top of the dashboard in large, colour-coded tiles so the reader grasps the headline in under five seconds.
- **2. Interactivity.** Slicers, timelines, and drop-downs let the user filter and drill down without leaving the sheet. A dashboard that cannot be filtered is just a report.

- **3. Visual hierarchy.** Most important charts are large and at the top; supporting details are smaller and below. Colour is used sparingly — one accent colour for emphasis, neutrals for context.
- **4. Clean, uncluttered layout.** Gridlines and headers are hidden; whitespace separates panels; chart titles are short. Every pixel should earn its place on the screen.
- **5. Mobile / print friendly.** The dashboard fits on a single screen at the target resolution (often 1366×768 for laptops, 1920×1080 for desktops). It also prints cleanly to a single A4 page for board packs.

Bonus features worth mentioning: (6) Automated data refresh via Power Query; (7) Conditional formatting that highlights exceptions; (8) A "last refreshed" timestamp so users know how stale the data is; (9) Hover-over cell comments explaining methodology; (10) A hidden "calculations" tab so the visible dashboard tab stays clean.

Python Programming Reference for DTS 104

This section consolidates the Python concepts that appear across the DTS 104 past questions: variables, lists, tuples, strings, slicing, loops, functions, and the built-in functions most useful for data analysis. Treat it as a quick-reference card while practising the programming sub-questions.

Lists — The Workhorse Data Structure

Lists are ordered, mutable collections. They are the default structure for storing a sequence of values in Python and are used in virtually every data-analysis script.

List operations cheat-sheet:

```
# Creation
empty = []
squares = [1, 4, 9, 16, 25]
mixed = [1, "two", 3.0, [4, 5]] # heterogeneous

# Indexing (0-based, negative indices count from end)
squares[0]      # 1      - first element
squares[-1]     # 25     - last element

# Slicing [start:stop:step] (stop is exclusive)
squares[1:4]    # [4, 9, 16]
squares[:3]     # [1, 4, 9]
squares[2:]     # [9, 16, 25]
squares[::2]    # [1, 9, 25] - every 2nd element
squares[::-1]   # [25, 16, 9, 4, 1] - reversed

# Mutation
squares.append(36)      # [1, 4, 9, 16, 25, 36]
squares.extend([49, 64]) # [1, 4, 9, 16, 25, 36, 49, 64]
squares.insert(0, 0)    # [0, 1, 4, 9, 16, 25, 36, 49, 64]
squares.remove(0)       # deletes first occurrence of 0
popped = squares.pop()  # removes and returns last element

# Inquiry
len(squares)            # 8
9 in squares            # True
squares.count(9)        # 1
squares.index(9)        # 2
```

For Loops and Enumerate

The for loop is the primary tool for iterating over a list, string, dictionary, or any iterable. `enumerate()` pairs each element with its index — essential when you need both.

For-loop patterns:

```
# Pattern 1 – simple iteration
scores = [85, 92, 78, 90, 88]
total = 0
for s in scores:
    total += s
print(f"Total = {total}, Mean = {total/len(scores):.2f}")

# Pattern 2 – enumerate (index + value)
for i, s in enumerate(scores, start=1):
    print(f"Student {i}: {s}")

# Pattern 3 – iterate two lists in parallel with zip()
names = ["Ade", "Bola", "Chidi"]
grades = ["A", "B", "C"]
for name, grade in zip(names, grades):
    print(f"{name} → {grade}")

# Pattern 4 – range with step
for n in range(0, 10, 2):    # 0, 2, 4, 6, 8
    print(n, end=" ")
```

Output:

```
Total = 433, Mean = 86.60
Student 1: 85
Student 2: 92
Student 3: 78
Student 4: 90
Student 5: 88
Ade → A
Bola → B
Chidi → C
0 2 4 6 8
```

Worked Example 1 — Compute Mean, Median, Mode Without Imports

Mean, median, mode from scratch:

```
def mean(xs):
    return sum(xs) / len(xs)

def median(xs):
    s = sorted(xs)
    n = len(s)
    mid = n // 2
    if n % 2 == 1:
        return s[mid]
    return (s[mid - 1] + s[mid]) / 2

def mode(xs):
    counts = {}
    for x in xs:
        counts[x] = counts.get(x, 0) + 1
    max_count = max(counts.values())
    return [x for x, c in counts.items() if c == max_count]

X = [5, 7, 3, 7, 10, 7, 4, 8, 6]
print(f"Mean    = {mean(X):.4f}")
print(f"Median  = {median(X)}")
print(f"Mode    = {mode(X)}")
```

Output:

```
Mean    = 6.3333
Median  = 7
Mode    = [7]
```

Worked Example 2 — Reverse and Slice a Nested List

Nested-list slicing:

```
data = [[1, 2], 3, [4, 5, 6, 7], [8, 9], 10]

# Outer-list operations
print("Length      :", len(data))
print("Slice [2:5]   :", data[2:5])
print("Reversed     :", data[::-1])

# Inner-list operations (e.g. the 3rd element which is [4,5,6,7])
inner = data[2]
print("\nInner [4,5,6,7]:")
print("  Length      :", len(inner))
print("  Slice [1:3]    :", inner[1:3])
print("  Reversed      :", inner[::-1])
```

Output:

```

Length      : 5
Slice [2:5]  : [[4, 5, 6, 7], [8, 9], 10]
Reversed    : [10, [8, 9], [4, 5, 6, 7], 3, [1, 2]]

Inner [4,5,6,7]:
  Length    : 4
  Slice [1:3] : [5, 6]
  Reversed   : [7, 6, 5, 4]

```

Worked Example 3 — Compute Variance and Standard Deviation

Variance and std-dev from scratch:

```

import math

def mean(xs):
    return sum(xs) / len(xs)

def p_variance(xs):      # population variance (divide by n)
    m = mean(xs)
    return sum((x - m) ** 2 for x in xs) / len(xs)

def s_variance(xs):      # sample variance (divide by n-1)
    m = mean(xs)
    return sum((x - m) ** 2 for x in xs) / (len(xs) - 1)

X = [5, 7, 3, 7, 10, 7, 4, 8, 6]
pvar = p_variance(X)
svar = s_variance(X)
print(f"Population variance = {pvar:.4f}")
print(f"Sample variance     = {svar:.4f}")
print(f"Population std dev   = {math.sqrt(pvar):.4f}")

```

Output:

```

Population variance = 4.0000
Sample variance     = 4.5000
Population std dev  = 2.0000

```

Worked Example 4 — Hero's Formula as a Reusable Function

Function with input validation:

```
import math

def heron_area(a, b, c):
    """Compute triangle area using Heron's formula."""
    # Triangle-inequality check
    if a + b <= c or a + c <= b or b + c <= a:
        raise ValueError(f"Sides {a}, {b}, {c} do not form a triangle.")
    s = (a + b + c) / 2
    return math.sqrt(s * (s - a) * (s - b) * (s - c))

# Test with three triangles
tests = [(3, 4, 5), (5, 12, 13), (7, 9, 12)]
for a, b, c in tests:
    print(f"Triangle ({a}, {b}, {c}) -> area = {heron_area(a, b, c):.4f}")
```

Output:

```
Triangle (3, 4, 5) -> area = 6.0000
Triangle (5, 12, 13) -> area = 30.0000
Triangle (7, 9, 12) -> area = 26.8328
```


Statistics for Data Analysis

Descriptive statistics summarise a dataset with a small set of numbers. They are the first thing an analyst computes after cleaning the data and the foundation on which all subsequent modelling rests. This section defines each measure, gives its formula, and works through the dataset $X = [5, 7, 3, 7, 10, 7, 4, 8, 6]$ used in Question 3(b).

Measures of Central Tendency

Measure	Formula	When to Use	Value for X
Mean (μ)	$\mu = \Sigma x / n$	Symmetric distributions, no extreme outliers.	$57 / 9 = 6.333$
Median	Middle value of sorted X (avg of two middle values if n is even).	Skewed distributions or data with outliers.	7
Mode	Most frequent value.	Categorical data; identifying the "typical" case.	7 (appears 3 times)

Table 9 — Measures of central tendency.

Measures of Dispersion

Measure	Formula	Interpretation	Value for X
Range	$\max(X) - \min(X)$	Total spread of the data; sensitive to outliers.	$10 - 3 = 7$
Variance (population)	$\sigma^2 = \Sigma(x - \mu)^2 / n$	Average squared deviation from the mean.	$36 / 9 = 4.000$
Variance (sample)	$s^2 = \Sigma(x - \bar{x})^2 / (n-1)$	Use when X is a sample (most real-world cases).	$36 / 8 = 4.500$
Standard deviation	$\sigma = \sqrt{\sigma^2}$	Same units as the data; interpretable spread.	$\sqrt{4} = 2.000$
Interquartile range (IQR)	$Q3 - Q1$	Spread of the middle 50%; robust to outliers.	$Q1=5, Q3=8, IQR=3$

Table 10 — Measures of dispersion.

Worked Example — Full Descriptive Summary in Python

Full descriptive statistics for $X = [5, 7, 3, 7, 10, 7, 4, 8, 6]$:

```

import statistics as st
import math

X = [5, 7, 3, 7, 10, 7, 4, 8, 6]

# Central tendency
mean_x = st.mean(X)
median_x = st.median(X)
mode_x = st.mode(X)

# Dispersion
range_x = max(X) - min(X)
var_pop = st.pvariance(X)
var_samp = st.variance(X)
std_pop = st.pstdev(X)
q1, q3 = st.quantiles(X, n=4)[0], st.quantiles(X, n=4)[2]
iqr = q3 - q1

print("=== Descriptive Statistics for X ===")
print(f"n : {len(X)}")
print(f"Mean : {mean_x:.4f}")
print(f"Median : {median_x}")
print(f"Mode : {mode_x}")
print(f"Range : {range_x}")
print(f"Variance (pop) : {var_pop:.4f}")
print(f"Variance (sample) : {var_samp:.4f}")
print(f"Std dev (pop) : {std_pop:.4f}")
print(f"Q1 : {q1}")
print(f"Q3 : {q3}")
print(f"IQR : {iqr}")

```

Output:

```

=== Descriptive Statistics for X ===
n : 9
Mean : 6.3333
Median : 7
Mode : 7
Range : 7
Variance (pop) : 4.0000
Variance (sample) : 4.5000
Std dev (pop) : 2.0000
Q1 : 5.0
Q3 : 8.0
IQR : 3.0

```

Choosing the Right Measure

- **Symmetric data, no outliers** → mean + standard deviation.
- **Skewed data or outliers present** → median + IQR. The mean is pulled toward outliers; the median is not.
- **Categorical data** → mode. Mean and median are not defined.
- **Ordinal data** → median + IQR. The mean is technically invalid because intervals between ranks are not necessarily equal.

Key Terms and Definitions

A condensed glossary of the terms that appear across the DTS 104 past questions. Memorise the definitions and the canonical example for each.

Term	Definition
Business Intelligence (BI)	The infrastructure, tools, and processes that collect, integrate, and present business data so that managers can monitor operations and discover opportunities.
Continuous Data	Quantitative data that can take any value within a range (height, temperature, revenue). Opposite of discrete data.
Dashboard	A single-screen visual display of the most important information needed to achieve one or more objectives, consolidated on a single screen for monitoring at a glance.
Descriptive Analysis	Analytics that summarises what happened in the past using aggregations, summaries, and visualisations.
Discrete Data	Quantitative data that can only take specific values, usually integers (number of students, number of cars).
Dispersion	The spread of values around the central value, measured by range, variance, standard deviation, or IQR.
Drop-down (Data Validation)	An Excel cell that shows a list of valid values when clicked, used to drive formula-driven KPIs and charts.
EDA (Exploratory Data Analysis)	Open-ended, hypothesis-free investigation of a dataset at the start of an analysis, combining summary statistics and visualisations.
Estimation	Inferring the value of an unknown population parameter from a sample, either as a point estimate or as a confidence interval.
Excel Dashboard	A single worksheet in Excel that combines KPI tiles, charts, and interactive filters to monitor a business process.
Hero's (Heron's) Formula	Formula for the area of a triangle from its three sides: $A = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = (a+b+c)/2$.
Interquartile Range (IQR)	The difference between the 75th and 25th percentiles ($Q3 - Q1$); a robust measure of spread.
List (Python)	An ordered, mutable, heterogeneous collection defined with square brackets; the workhorse data structure for sequential data.
Mean (μ)	The arithmetic average of a dataset, computed as $\Sigma x / n$.
Median	The middle value of a sorted dataset; robust to outliers.
Mode	The most frequently occurring value in a dataset.
Nominal Data	Qualitative data with no inherent order (eye colour, nationality).

Term	Definition
Ordinal Data	Qualitative data with a meaningful order but unequal intervals (education level, satisfaction rating).
Outlier	An observation that lies an abnormal distance from other values; commonly defined as more than $1.5 \times \text{IQR}$ below Q1 or above Q3.
Predictive Analysis	Analytics that forecasts likely future outcomes using statistical or machine-learning models.
Prescriptive Analysis	Analytics that recommends the best action to take, often using optimisation or simulation.
Qualitative Data	Non-numerical data describing categories or qualities; sub-types are nominal and ordinal.
Quantitative Data	Numerical data representing counts or measurements; sub-types are discrete and continuous.
Slicer (Excel)	A graphical filter panel with clickable buttons that connects to one or more PivotCharts and updates them simultaneously.
Standard Deviation (σ)	The square root of variance; same units as the original data.
Structured Data	Data organised into a fixed schema of rows and columns; stored in relational databases or spreadsheets.
Timeline (Excel)	A date-specific slicer with a draggable slider for filtering by month, quarter, or year.
Trend Analysis	Examination of data over time to identify the direction and rate of change; uses moving averages, regression, or decomposition.
Tuple (Python)	An ordered, immutable, heterogeneous collection defined with parentheses; used for fixed records.
Unstructured Data	Data with no pre-defined schema (text, images, audio, video); requires NLP, OCR, or computer vision to analyse.
Variance (σ^2)	The average of the squared deviations from the mean; measures spread.
Visualizing Data	The strategy of representing numerical data graphically so that patterns, relationships, and outliers become immediately visible.

Table 11 — Glossary of DTS 104 key terms.

Summary and Quick Review

Use this page as a final revision sheet the night before the exam. Each item is one sentence — the absolute minimum you must remember.

Concepts in One Line Each

- **Data analysis** = systematic transformation of raw data into actionable insight.
- **Structured vs unstructured** = schema'd tables vs free-form text/media (20% vs 80% of world data).
- **Descriptive / Predictive / Prescriptive** = what happened / what will happen / what to do.
- **Qualitative vs Quantitative** = categories vs numbers. Nominal+ordinal vs discrete+continuous.
- **Mean** = $\Sigma x / n$ | **Median** = middle of sorted X | **Mode** = most frequent.
- **Variance (pop)** = $\Sigma (x - \mu)^2 / n$ | **Sample** = $\Sigma (x - \bar{x})^2 / (n - 1)$. **Std dev** = $\sqrt{\text{variance}}$.
- **EDA** = open-ended exploration with summary stats + plots, before modelling.
- **Excel cleaning** = Find > Blanks, Conditional Formatting > Duplicates, Remove Duplicates, AVERAGE/MEDIAN imputation.
- **Excel dashboard** = KPIs on top, slicers + timelines for interactivity, line/combo charts for trends, clean layout.
- **Python list** = ordered, mutable, heterogeneous. **Tuple** = immutable. **String** = immutable characters.
- **Slicing** = data[start:stop:step], stop exclusive, data[::-1] reverses.
- **Hero's formula** = $\sqrt{s(s-a)(s-b)(s-c)}$, $s = (a+b+c)/2$.
- **5 tools** = Python, R, Excel, SQL, Tableau/Power BI.
- **Data team roles** = Analyst, Scientist, Engineer, BI Developer, Domain Expert.

Key Formulas

Formula	Use
$\mu = \Sigma x / n$	Population mean
$s = (a + b + c) / 2$	Semi-perimeter of a triangle
$A = \sqrt{s(s-a)(s-b)(s-c)}$	Heron's formula for triangle area
$\sigma^2 = \Sigma (x - \mu)^2 / n$	Population variance
$s^2 = \Sigma (x - \bar{x})^2 / (n - 1)$	Sample variance
$\sigma = \sqrt{\sigma^2}$	Standard deviation
$IQR = Q3 - Q1$	Interquartile range
data[start:stop:step]	Python list slicing

Table 12 — Formula card.

Five Rapid-Fire Self-Test Questions

- **Q1.** What are the seven steps of the data-analysis pipeline? (Collect → Clean → EDA → Transform → Model → Interpret → Communicate.)
- **Q2.** Compute mean, median, mode of [4, 4, 6, 8, 10]. (Mean 6.4, Median 6, Mode 4.)
- **Q3.** What does `data[2:5]` return for `data = [10, 20, 30, 40, 50, 60]`? ([30, 40, 50].)
- **Q4.** Name three chart types suitable for showing trends. (Line, column, combo.)
- **Q5.** What is the difference between a slicer and a drop-down? (Slicer = visual filter panel that drives many charts; drop-down = single-cell list that drives formulas.)

End of DTS 104 study guide. Best of luck in your examination.